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THE HISTORY OF

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SMALLTALK

Message-passing, children, and the invention of object-oriented personal computing at Xerox PARC

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STARTING POINT: THE EARLY HISTORY OF SMALLTALK

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The definitive first-person account of Smalltalk is Alan Kay's "The Early History of Smalltalk," written for the second ACM History of Programming Languages conference (HOPL II, 1993). This whitepaper takes as its starting point the annotated, reader-repaired hypertext edition maintained by Bret Victor at worrydream.com [1] -- itself an artifact of the hypertext tradition covered elsewhere in this series -- and the original ACM publication [2]. A vendored copy of that edition, including Kay's scanned figures, travels with this repository in TheHistoryOf/EarlyHistoryOfSmalltalk.

Kay's essay frames Smalltalk not as a programming language project but as a medium: "a new medium for communication," whose design fell out of a question about what computers could be for children. Everything below is a compressed map of that story and what happened after it.

1960S ROOTS: SKETCHPAD, SIMULA, THE FLEX MACHINE

Three streams feed Smalltalk. Ivan Sutherland's Sketchpad (1963) demonstrated that a display could hold live, constrained, manipulable objects -- "the first object-oriented graphics system" in spirit. Simula (Dahl and Nygaard, 1962-67) carried the idea of classes of activity records into ALGOL's world; Kay encountered it as a graduate student at Utah in 1966 and read its tape listings until the idea clicked: each object could be a little computer. And Kay's own FLEX machine (with Ed Cheadle, 1967-69) tried to build a personal computer for nonexperts, whose failures -- it was, in Kay's words, too much like a "shotgun marriage" of ideas -- taught him what a truly simple, uniform system would have to feel like.

The fourth stream was biological. Kay, a former biology major, kept returning to the cell as the model: thousands of quasi-autonomous units communicating by messages, with the complexity hidden behind membranes. "The big idea is messaging," he would later insist -- not classes, not inheritance.

PARC AND THE CHILDREN

In 1970 Xerox founded the Palo Alto Research Center, and Kay's Learning Research Group set out to build the interim Dynabook (see the Dynabook chapter): hardware called the Alto, and a software medium for it. Smalltalk-71 was a paper design; Smalltalk-72 became real after Dan Ingalls implemented, in BASIC and then assembly, a scheme Kay had bet he could fit on one page. Everything was an object; computation was objects sending messages; children in Palo Alto schools -- Adele Goldberg's studies with twelve-year-olds are the famous ones -- built drawing tools, animations, and music systems in it.

Smalltalk-76 (Ingalls) added classes, inheritance, and the

bytecode virtual machine shape that survives today. The overlapping-window, icon, menu interface built to surround it became, by way of the famous 1979 Steve Jobs visit, the direct ancestor of the Lisa, the Macintosh, and every desktop since. Kay's talk "Doing with Images Makes Symbols" [5] narrates this whole UI lineage from Sketchpad and NLS through Smalltalk.

SMALLTALK-80 AND RELEASE TO THE WORLD

Smalltalk-80 was the cleaned-up, portable system Xerox finally released. The August 1981 issue of BYTE magazine [3] -- the one with the hot-air balloon leaving Smalltalk island on the cover -- introduced it to the industry, with articles by Goldberg, Ingalls, Robson, and the rest of the Learning Research Group. The "Blue Book" (Goldberg and Robson, Smalltalk-80: The Language and its Implementation, 1983) [4] specified the virtual machine so exactly that new implementations could be, and repeatedly were, bootstrapped straight from its pages -- the property that later made Squeak possible (see the Squeak chapter).

Commercial descendants (ParcPlace's ObjectWorks/VisualWorks, Digitalk, IBM VisualAge) carried Smalltalk through the late 1980s and 1990s. Its ideas -- objects, messages, the class browser, the debugger you can program in, refactoring, unit testing (Kent Beck's SUnit begat JUnit and the xUnit family), model-view-controller -- leaked into essentially every subsequent language and IDE, usually with the dynamism filed off. Kay's HOPL essay ends with a warning that remains the series' theme: the computer revolution hasn't happened yet.

SOURCES & CITATIONS

- [1] Alan Kay, The Early History of Smalltalk, annotated
hypertext edition maintained by Bret Victor

<https://worrydream.com/EarlyHistoryOfSmalltalk/>

Vendored in this repo at

[TheHistoryOf/EarlyHistoryOfSmalltalk](#) (fork of github.com/worrydream/EarlyHistoryOfSmalltalk).

- [2] Alan C. Kay, The Early History of Smalltalk, HOPL II, ACM, 1993

<https://dl.acm.org/doi/10.1145/155360.155364>

- [3] BYTE Magazine vol. 6 no. 8, August 1981: the Smalltalk-80 issue, on the Internet Archive

<https://archive.org/details/byte-magazine-1981-08>

- [4] Adele Goldberg and David Robson, Smalltalk-80: The Language and its Implementation (the Blue Book), free scan via Stephane Ducasse's collection of Smalltalk books

<http://stephane.ducasse.free.fr/FreeBooks.html>

- [5] Alan Kay, Doing with Images Makes Symbols: Communicating with Computers (1987), Internet Archive

https://archive.org/details/Dr._Alan_Kay_Doing_With_Images_Makes_Symbo

- [6] Ivan Sutherland, Sketchpad: A Man-Machine Graphical Communication System (1963); demo footage survives in Alan Kay's 1987 lecture above

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